

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 07

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>Correct answer only.</i> • B - Steam washing (hot water / steam jets kill organisms in the sediment) (1)	1
01.2	<i>1 mark for each point.</i> • Chemicals cause the oil molecules to join together / solidify (1) • Producing more solid material that can be collected / lifted more easily than liquid oil (1)	2
01.3	<i>1 mark for each point.</i> • Air contains about 210 cm ³ of oxygen per litre whereas water at 20 °C holds a maximum of about 6 cm ³ per litre (less than 3% of the amount in air) (1) • So a small amount of oxygen-consuming activity (decomposition, respiration) or warming can remove most of the oxygen; many aquatic organisms are near the lower end of their range of tolerance (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 254-262; AQA spec 3.4.3.2.6-3.4.3.2.7.	
02.1	<i>1 mark for the correct answer.</i> • $(44 - 8) / 50 = 0.72 \text{ m yr}^{-1}$ (1)	1
02.2	<i>1 mark for each point.</i> • Abstraction (for irrigation) exceeds the rate of natural recharge by infiltration / percolation of precipitation (1) • The dynamic equilibrium between inflow and outflow has been upset so the stored volume declines (1)	2

Question	Marking guidance	Marks
02.3	<i>1 mark for each point.</i> • Porosity: a high proportion of the rock volume is space that can hold water (chalk, limestone, sandstone) (1) • Permeability: the spaces are interconnected and large enough for water to flow through (clay is porous but not permeable) (1) • Impermeable rock (granite, clay) beneath to stop water escaping, and permeable rock / soil above to allow recharge (1)	3
02.4	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Salt water incursion: seawater flows into the aquifer to replace the fresh water, making it unusable for irrigation (osmotic dehydration of crops) (1) • Subsidence: emptied pore spaces no longer support the rock so it compacts, damaging buildings and pipes (eg Mexico City) (1) • Drying up of groundwater-fed rivers, lakes and wetlands as the water table falls below their beds (1) • Ecological impacts: plants needing high water availability die; wetland animals lost; food chain effects (1) • Reduced supplies: abstraction may suddenly have to stop after years of hidden decline (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 119-121; AQA spec 3.2.2.1-3.2.2.2.	
03.1	<i>Correct answer only.</i> • B - tiny pores between clay particles allow capillary rise; none in sand (1)	1
03.2	<i>1 mark for each point.</i> • Hygroscopic: held tightly by adhesive forces on particle surfaces; cannot drain and cannot be absorbed by plants (1) • Capillary: held in pores by surface tension; does not drain but can be absorbed by plants (1) • Gravitational: drains freely through large pores; available to plants only briefly as it passes (1)	3
03.3	<i>1 mark.</i> • Small round crumbs leave spaces giving good drainage, aeration and easy root penetration; large flat platy peds pack together reducing all three (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-169; AQA spec 3.2.5.	
04.1	<i>1 mark for each point.</i> • Wheel vibration: track polishing / grinding, sound-absorbing ballast, sound-absorbing suspension (1) • Pantograph: aerodynamic fairing (1) • Wheel squeal: lubrication of wheels / track on corners (1) • Braking: composite-material brakes instead of cast iron (1)	4
04.2	<i>1 mark.</i> • Roads pass through / near almost all residential areas and traffic is continuous through the day, unlike airports or railways which affect limited areas (1)	1

Question	Marking guidance	Marks
04.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Sound-absorbing road surfaces (porous asphalt / rubber-crumb asphalt) (1) • Acoustic fences, embankments / baffle mounds, walls or dense tree planting (1) • Traffic management for free flow (less acceleration noise); re-routing traffic away from residential areas (1) • Double glazing / acoustic insulation of buildings (1) • Vehicle design: better aerodynamics, engine insulation, quieter exhausts, quieter tyres (1)	3
04.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Air bubble curtains around the pile driver absorb noise (1) • Acoustic survey to detect cetaceans first and delay work if present; increase volume gradually (soft start) so animals can leave (1) • Drilling instead of pile-driving (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 292-294; AQA spec 3.4.3.2.13.	
05.1	<i>Correct answer only.</i> • C (1)	1
05.2	<i>1 mark for each point.</i> • Monitoring can be focused on the sites predicted to be most at risk / where the pollutant will concentrate (1) • No need to sample sites that were never likely to be polluted; sampling also checks the accuracy of the predictions (1)	2
05.3	<i>1 mark for each point.</i> • Behaviour of new chemicals can be predicted from similar existing chemicals: it is likely to persist, bioaccumulate and biomagnify (1) • So acceptable emission limits, control methods and monitoring programmes can be set before release (precautionary principle) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245; AQA spec 3.4.3.1.	
06.1	<i>1 mark for each point.</i> • UV light splits diatomic oxygen (O₂) into two monatomic oxygen atoms (O) (1) • Monatomic oxygen reacts with diatomic oxygen to form ozone (O₃) (1) • UV light also splits ozone into O₂ and O; the reactions form a dynamic equilibrium which absorbs UV (converting it to chemical energy / heat) (1)	3
06.2	<i>Up to 6 marks in total; award marks in proportion to the completeness of each point.</i> • Useful: boiling points near ambient temperature so easily liquefied under low pressure (aerosol propellants, fridge / air-conditioning coolants); dissolve grease but do not damage electrical components (solvents); non-flammable; non-toxic (up to 2) • Persistence: chemically stable / unreactive in the troposphere so they remain long enough to be carried up to the stratosphere (up to 2) • Dissociation by UV: in the stratosphere higher UV levels break the carbon-chlorine bonds releasing chlorine free radicals (up to 2) • Chlorine reacts with monatomic oxygen (and ozone), preventing ozone formation; chlorine is regenerated so one atom destroys many ozone molecules; bromine / iodine act similarly (up to 2)	6

Question	Marking guidance	Marks
06.3	<i>1 mark for each point.</i> • UVB energy is absorbed by cells and breaks up biological molecules / DNA causing mutations, skin cancer, cataracts, sunburn (human health) (1) • Damage to leaf tissue and reduced photosynthesis / crop yields (plants) (1) • Damage to marine algae, phytoplankton, corals and planktonic larvae (marine organisms), reducing productivity and food webs (1)	3
06.4	<i>1 mark.</i> • An area where the total ozone column falls below 220 Dobson Units (normal is about 300 DU) - a level never recorded before CFC pollution (1)	1
06.5	<i>1 mark.</i> • Concentration varies with altitude and the stratosphere is not normally sampled; Dobson Units measure the total column thickness instead (1)	1
06.6	<i>1 mark.</i> • The British Antarctic Survey at Halley detected higher UV at ground level, showing ozone above had decreased (satellites above the stratosphere cannot measure downward UV directly) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-109; AQA spec 3.2.1.3.	
07.1	<i>1 mark for each point.</i> • Most solar energy is absorbed at the surface in the tropics where insolation is most intense / the Sun is overhead (1) • The warm surface heats the air above; warm winds (eg south-westerlies) carry heat energy to higher latitudes (1) • Winds blowing over the oceans create currents (North Atlantic Conveyor / Gulf Stream) carrying warm water from the tropics to NW Europe (1) • Evaporation in the tropics stores latent heat in water vapour which is released when it condenses at higher latitudes; also transports water to areas that would be dry (1)	4
07.2	<i>1 mark for each point.</i> • The North Atlantic Conveyor brings warm water from the tropics to the coast of NW Europe (1) • Prevailing south-westerly winds blowing over the warm sea carry the heat onto the land (1)	2
07.3	<i>1 mark for each point.</i> • Atmospheric pressure controls how easily water molecules evaporate / escape from the surface (1) • At low pressure water would boil / evaporate at low temperatures so all surface water would become vapour (1)	2
07.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Nitrogen (1) • Oxygen (1) • Carbon dioxide (1) • Argon / neon / krypton / xenon (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 87, 96-97; AQA spec 3.2.1.1, 3.2.2.3.	

Question	Marking guidance	Marks
08.1	<i>1 mark for each point.</i> - Sieves assembled with the coarsest mesh on top; a dried, crushed soil sample is placed in the top sieve (1) - Stack shaken (about 2 minutes) until no more particles pass; each fraction weighed (1) - Percentages of sand (0.02-2 mm), silt (0.002-0.02 mm) and clay (base pan, below 0.002 mm) calculated; material over 2 mm (stones, twigs) is not soil; plotted on a soil triangle (1)	3
08.2	<i>1 mark for each point.</i> - Auger collects a core of known cross-sectional area and depth (1) - So samples from different sites are directly comparable (also needed for bulk density = dry mass / volume) (1)	2
08.3	<i>1 mark for each point.</i> - Minerals such as calcium carbonate break down releasing carbon dioxide (1) - So mass is lost that is not organic matter (1)	2
08.4	<i>1 mark for each point.</i> - A beam of light passes through the sample in a dark chamber; a light sensor at 90° measures light scattered by suspended particles - more turbid, more scatter (1) - Standardise bulb brightness / colour, distance and container (1) - A calibration curve is produced using samples with known amounts of suspended matter so readings convert to concentrations; Secchi discs / turbidity bottles give only semi-quantitative results (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 410-413, 417; AQA spec 3.7.2.1-3.7.2.2.	
09.1	<i>1 mark for the correct answer.</i> $180 / 1.2 = 150$ (1)	1
09.2	<i>1 mark.</i> Cadmium and zinc occur together in the same ores so mine drainage / smelting releases both (1)	1
09.3	<i>1 mark for each point.</i> - Mine drainage water contaminated irrigation water / paddy soil so rice absorbed cadmium (1.2 vs 0.05 mg kg^{-1}) (1) - Cadmium is liposoluble / not excreted so it bioaccumulates in the body (kidneys) with chronic exposure to small doses over years (1) - Cadmium causes kidney failure and bone decalcification / skeletal collapse (itai-itai disease) (1) - Rice is the staple food so residents receive a large dose; village B has no exposure and no disease (correlation supports causation) (1)	4
09.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> - Collect mine drainage water and neutralise / raise pH with lime so cadmium precipitates and stays insoluble (1) - Cover / cap spoil heaps to prevent leaching and dust (1) - Recycle nickel-cadmium batteries / hazardous waste landfill (1) - Phytoremediation of contaminated paddy soils (1)	2
09.5	<i>1 mark for each point.</i> - Manufacture may release cadmium in effluents / dust (1) - Panels have a limited life; disposal of old panels (landfill, incineration of pigments) could release cadmium unless recycled (1)	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-279; AQA spec 3.4.3.2.11.	
10.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • No reservoir range = $3.9 - 0.7 = 3.2$ m; with reservoir = $2.7 - 1.7 = 1.0$ m (1) • Reduction = 2.2 m (1)	2
10.2	<i>1 mark for each point.</i> • After heavy rainfall surplus water is stored, reducing peak flows / flooding downstream (1) • During dry periods the dam is opened to release water and maintain river levels (1)	2
10.3	<i>1 mark for each point.</i> • Deforestation / urbanisation reduce infiltration and interception (1) • So runoff reaches the river faster after rain and less water is stored to maintain flow in dry periods (1)	2

Question	Marking guidance	Marks
10.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Site factors: topography - narrow exit from a large deep basin so a small dam holds a large volume; impermeable, strong bedrock without faults or seismic activity; large catchment with regular high rainfall and low evaporation; catchment land use with low pollution risk (pesticides, industry) and low soil erosion to avoid sedimentation; forest / vegetation in the flooded area decaying anaerobically releasing methane; access for workers and materials and distance to demand; value of the land lost (urban, conservation, agricultural) - loss-benefit analysis differs between countries • Alternatives for supply: rainwater collection, rivers, estuary barrages, desalination (energy-intensive so only where freshwater is inadequate), unexploited aquifers (N Africa, Kenya), inter-basin transfers (China, Australia, Wales-England), artificial aquifer recharge using surplus wet-season water, river-regulation reservoirs, afforestation to even out flow • Conservation strategies: low-water appliances, dual-flush toilets, xeriscaping, drip irrigation, grey-water re-use, leak repair (about 20% lost in the UK), metering • Treatment / pollution control allowing re-use of contaminated sources: sewage treatment, buffer strips, mine drainage control • Evaluation: reservoirs give large reliable supplies and flood control but flood land, cause habitat loss and methane; conservation is cheap and quick but limited by behaviour; transfers create conflicts between regions; desalination has high energy / carbon cost; combination needed as demand grows • Justified conclusion	9
	Model answer - Level 3 (8 marks)	
	A reservoir site is judged first on topography: a narrow exit from a large, deep basin lets a small, cheap dam hold a large volume. The bedrock must be impermeable so water is not lost and strong enough, without faults or seismic activity, to carry the dam. The catchment should be large with regular, high rainfall and low evaporation, and its land uses should not pose a pollution risk from industry or pesticides or cause the soil erosion that would silt the reservoir. Flooding forest or vegetation causes anaerobic decay and methane. Access for workers and materials, distance to the area of demand, and the value of the land lost - urban areas and nature reserves protected, farmland less so - complete a loss-benefit analysis that different countries weigh differently.	

Question	Marking guidance	Marks
	Supply can also be increased without new reservoirs: rainwater collection in cities, estuary barrages, desalination where freshwater is genuinely inadequate, newly found aquifers in North Africa and Kenya, inter-basin transfers as in China and from Wales to England, artificial recharge of aquifers with wet-season surplus, river-regulation reservoirs and afforestation to even out flow. Demand can be cut by low-water appliances, xeriscaping, drip irrigation, grey-water re-use, metering and repairing the leaks that lose about a fifth of UK supply, and treatment lets contaminated sources be re-used.	
	Reservoirs give large, reliable supplies and flood control but flood land and habitats and release methane; conservation is cheap and quick but limited by behaviour; transfers create conflicts between regions; desalination carries a high energy and carbon cost. As demand grows, the sustainable answer is conservation first, then a combination of sources matched to each region.	
	Weaker answer - Level 2 (4 marks)	
	A reservoir needs a large catchment area with lots of rain, impermeable rock so the water does not drain away, and a narrow valley so the dam is small. It should be near the people who need the water and not flood important land. Other ways to get water include desalination, which is expensive, and collecting rainwater. Water can be conserved by fixing leaks, using water meters and having low-flush toilets. All of these help make water supplies more sustainable.	
	What makes the difference	
	• Level 2 lists correct site factors but without the reasons (why impermeable, why the valley shape matters, why land use in the catchment matters, methane from flooded vegetation).	
	• Level 3 covers far more alternatives (aquifer recharge, river regulation, transfers, afforestation) and evaluates each; Level 2 evaluates only desalination.	
	• Level 3 ends with a prioritised judgement; Level 2 ends with 'all of these help'.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-127; AQA spec 3.2.2.4-3.2.2.5.	

Question	Marking guidance	Marks
11.1	Discuss the strategies that can be used to provide sustainable supplies of water for a growing population. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Context: uneven distribution of water, growing population and per capita demand, unsustainable exploitation of rivers and aquifers, hydrological cycle links • New sources: rainwater collection, rivers (discharge, fluctuations, contaminants), reservoirs and site selection, estuary barrages, desalination, unexploited aquifers • Management: artificial aquifer recharge, river-regulation reservoirs, inter-basin transfers, afforestation • Conservation: low-water appliances, xeriscaping, drip irrigation, grey water, leak reduction, metering; pollution control to protect sources • Treatment: matching quality to use; screens, sedimentation, aeration, flocculation, filtration, sterilisation, pH adjustment, activated carbon, reverse osmosis • Evaluation of costs, energy, environmental impacts (habitat loss, methane, inter-tidal habitats, conflicts) and justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-127; AQA spec 3.2.2.4-3.2.2.5.	25

Question	Marking guidance	Marks
11.2	Discuss how the properties of solar power and hydroelectric power affect their use, and evaluate how new technologies could increase their contribution to energy supplies. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Problematic properties of solar: intermittency, unreliability from cloud, low energy density, changing angle; locational constraints (deserts, latitude) • Harnessing solar: photothermal, thermal stores, passive architecture, heat pumps, PV principles and cell efficiencies, solar farms; environmental impacts of manufacture and land use • New solar technologies: multijunction, anti-reflective surfaces, CSP with molten salt, PV-thermal hybrids, transparent PV, heliostats, self-cleaning • HEP: gravitational potential energy, locational constraints (catchment, rainfall, turbidity, impermeable rock, seismic, topography, land use, distance), impacts on habitats, sediment, methane, communities • Energy storage and matching supply to demand: pumped storage, batteries, grid links; combining resources • Evaluation and justified conclusion on how far these resources can replace fossil fuels Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 196-203, 217-224; AQA spec 3.3.4.1.	25